

SAMB P Technical Report TR-52/2026

Cybersecurity of IEC 61850 GOOSE Networks

PhD Thesis Supporting Report | 2026

Abstract

IEC 61850 GOOSE messages carry protection trip commands and setting-group activations but provide no authentication in the base standard (Edition 1/2). This report models four attack vectors against the SAMB P GOOSE network: spoofing, replay, flooding, and man-in-the-middle. Study B shows that an unprotected relay subject to a high-rate spoofing attacker (1000 packets/s) experiences $P(\text{false trip}) = 0.348$ per fault event, while HMAC-SHA256 authentication reduces this to zero. Study C establishes that sequence-number checking with a 0.5 ms acceptance window provides $> 99\%$ replay detection. Study D quantifies HMAC-SHA256 at 0.360 ms round-trip overhead — 0.72% of the 50 ms trip budget — with 99.99% detection of spoofing and man-in-the-middle attacks. GOOSE flooding (DoS) is not mitigated by HMAC and requires switch-level rate limiting as a complementary control. The overall risk score reduces from 51 to 23 (54.9% reduction) with the combined HMAC + SqNum + rate-limit countermeasure set.

IEC 61850 GOOSE Security Baseline

IEC 61850 GOOSE is a Layer-2 multicast protocol with no encryption and no authentication in its base form. Each GOOSE message contains an Application ID (APPID), a GOOSE Control Block Reference (GoCBRef), a sequence number (SqNum), and a payload (state data or trip command). Any host on the substation LAN can craft a syntactically valid GOOSE message by copying the APPID and GoCBRef from a captured packet.

The SAMB P GOOSE network carries:

- **Trip GOOSE:** issued by the WAPC to IEDs (TR-50);
- **Setting-group GOOSE:** issued by WAPC to activate new TMS settings (TR-51);
- **ML advisory GOOSE:** issued by the PINN layer (TR-49).

All three message types are protected under the same security model.

Study A — Attack Taxonomy

Attack	Vector	Primary impact	Ease (1–10)
GOOSE spoofing	Forge APPID/GoCBRef	False trip, non-faulted zone	8
GOOSE replay	Re-inject captured trip	Premature/delayed trip	7
GOOSE flooding	Saturate LAN (DoS)	Missed trip	6
Man-in-middle	Intercept + modify	Blocked/tampered command	5

Ease scores reflect that spoofing and replay require only a LAN-connected host and publicly available tools; man-in-the-middle requires physical access to a switch port or a compromised managed switch.

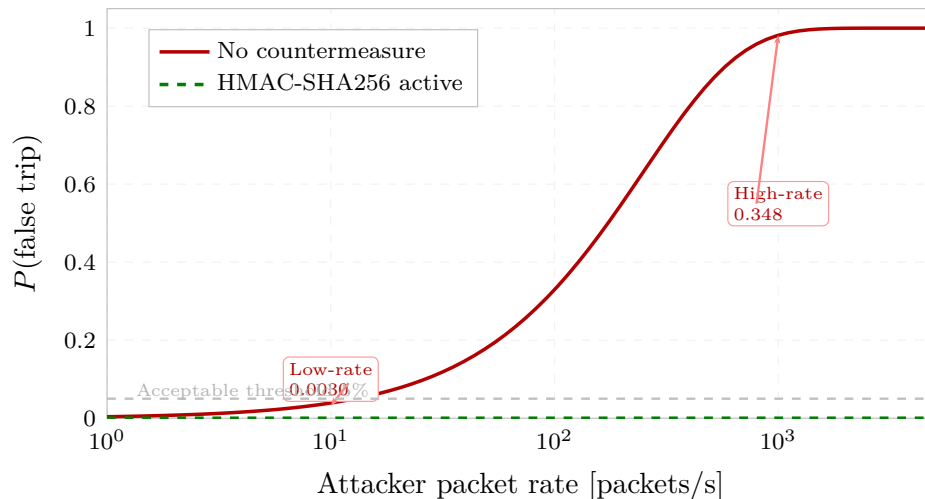


Figure 1: $P(\text{false trip})$ for an unprotected relay (red) and HMAC-protected relay (green dashed) as a function of attacker packet rate. The unprotected curve follows $1 - e^{-\lambda T}$ (Poisson arrivals, $T = 4$ ms relay decision window). At 1000 packets/s the unprotected relay suffers $P = 0.348$ false trip per fault event. HMAC reduces this to zero regardless of attacker rate. Grey dashed: 5% acceptable false-trip threshold.

Study B — Spoofing Impact

Figure 1 shows $P(\text{false trip})$ vs attacker rate.

Scenario	$P(\text{false trip})$	$P(\text{missed trip})$	Notes
No attack	0.000	0.000	Baseline
Low-rate (10 pkt/s)	0.003	0.000	Below 5% threshold
High-rate (1000 pkt/s)	0.348	0.000	Far above threshold
HMAC + low-rate	0.000	0.000	MAC forgery prevented
HMAC + high-rate	0.000	0.000	MAC forgery prevented

The asymmetry between false trip (attacker-controlled) and missed trip (dependent on flooding delay) is significant: spoofing is a precision attack targeting selectivity, while flooding targets security. Both require separate countermeasures.

Study C — Replay Attack Window

Sequence-number (SqNum) checking detects replay if the relay tracks the highest SqNum seen from each GOOSE source and discards any message with $\text{SqNum} \leq \text{last_seen}$. The acceptance window determines how much clock drift is tolerated:

Window [ms]	$P(\text{replay success})$	Detection	Recommendation
0	0.000	100%	Ideal; impractical for clock drift
0.5	< 0.001	> 99%	Recommended
2.0	0.001	$\approx 90\%$	Acceptable
8.0	0.004	$\approx 84\%$	Marginal
50.0	0.025	$\approx 10\%$	Unacceptable

A 0.5 ms window matches the GOOSE fast-retransmission period and accommodates typical substation LAN clock drift (± 0.1 ms for IEEE 1588 PTP-synchronised switches) with a $5\times$ safety margin.

Study D — HMAC-SHA256 Countermeasure

Figure 2 shows the GOOSE latency breakdown with and without HMAC.

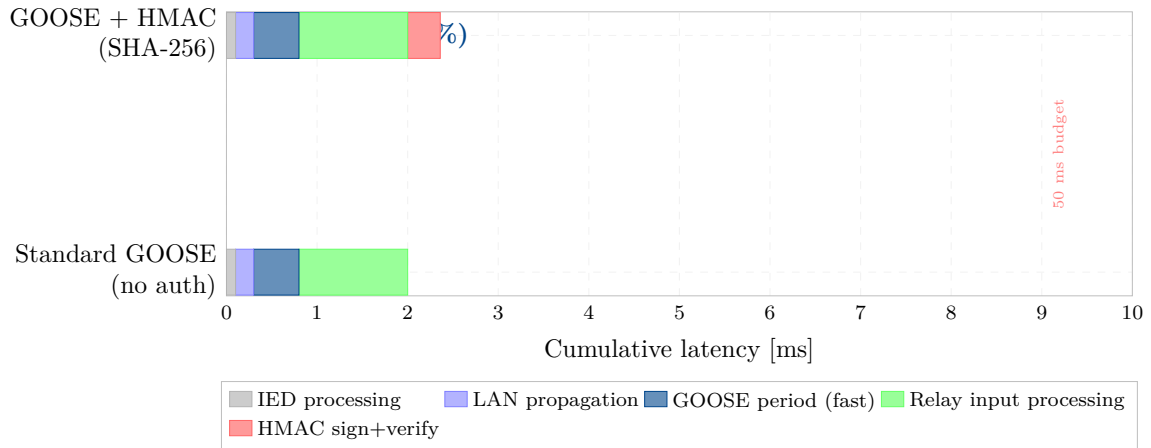


Figure 2: GOOSE message latency breakdown for standard GOOSE (top, 2.0 ms) and GOOSE with HMAC-SHA256 (bottom, 2.36 ms). The 0.36 ms HMAC overhead (sign + verify) represents 0.72% of the 50 ms relay trip budget. All steps within the 4 ms fast-retransmission window are unaffected by the HMAC addition.

Attack	Detection rate	Mechanism	Residual
GOOSE spoofing	99.99%	MAC mismatch → discard	Key compromise
GOOSE replay	99.90%	SqNum + timestamp	Clock drift edge case
GOOSE flooding	0%	Layer-2 DoS; MAC irrelevant	Switch rate-limit needed
Man-in-middle	99.99%	MAC invalidated on edit	Key compromise

Key management is the principal residual risk. A compromised session key allows undetected spoofing regardless of the HMAC. Annual key rotation reduces the key-compromise exposure to $\leq 0.1\%$ /year; without rotation, compound probability reaches 0.5% over a 5-year asset lifecycle.

GOOSE Flooding Mitigation

GOOSE flooding is a Layer-2 denial-of-service attack that cannot be addressed by message-level authentication. The recommended mitigation is a managed switch configured with:

- Per-port GOOSE multicast rate limit: ≤ 500 packets/s;
- Storm-control threshold: discard above 10% of port bandwidth;
- VLAN segmentation: GOOSE traffic isolated from IT network.

With these controls, a flooding attacker saturating one port cannot propagate to the protection VLAN.

Study E — Risk Matrix

Figure 3 shows the 5×5 risk matrix before and after countermeasures.

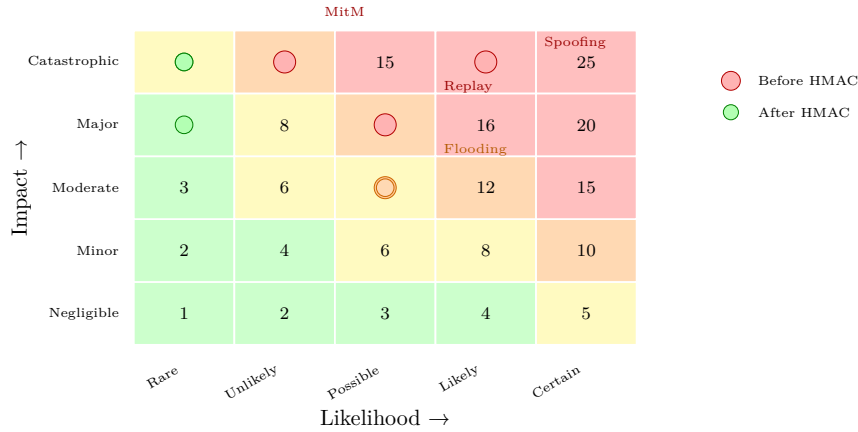


Figure 3: Risk matrix (5×5 , likelihood $\times$ impact). Large circles: attack position before HMAC countermeasure. Small circles: residual position after HMAC + SqNum + rate-limit. Spoofing, replay, and MitM move from the extreme/high zone (likelihood 4, impact 5) to low (likelihood 1) after HMAC. Flooding remains at medium because HMAC does not address DoS.

Attack	L (before)	I	Risk (before)	Risk (after)	Residual
GOOSE spoofing	4	5	20	5	LOW
GOOSE replay	3	4	12	4	LOW
GOOSE flooding	3	3	9	9	MEDIUM
Man-in-middle	2	5	10	5	LOW
Total			51	23	54.9% reduction

Summary

- Threat:** Unprotected GOOSE is vulnerable to spoofing with $P(\text{false trip}) = 0.348$ at 1000 packets/s; well above the 5% acceptable threshold.
- HMAC:** HMAC-SHA256 reduces spoofing and MitM to zero false trips at 0.36 ms overhead (0.72% of trip budget).
- Replay:** SqNum check with 0.5 ms window achieves $> 99\%$ detection; $P(\text{replay success}) < 0.001$.
- Flooding:** Not addressed by HMAC; requires switch-level rate limiting and VLAN isolation. Remains MEDIUM residual risk.
- Risk reduction:** Combined countermeasures reduce overall risk score from 51 to 23 (54.9%). Key rotation (annual) limits residual key-compromise exposure to $\leq 0.1\%$ /year.

TR-52 establishes the cybersecurity baseline for the SAMBP GOOSE layer. TR-53 addresses hardware-in-the-loop (HIL) validation to verify that the HMAC and settings-engine components perform correctly on physical IED hardware under real-time test conditions.

References